

$$(13) (i) \lim_{x \rightarrow 2} \frac{x-2}{\sqrt{x+2}} \quad (ii) \lim_{x \rightarrow \infty} \left(2 - \frac{1}{x} + \frac{4}{x^2} \right)$$

$$\lim_{x \rightarrow 2} \frac{x-2}{\sqrt{x+2}}$$

$$\lim_{x \rightarrow 2} \sqrt{x+2}$$

$$\frac{0}{2} = 0$$

$$\lim_{x \rightarrow \infty} 2 - \lim_{x \rightarrow \infty} \frac{1}{x} + \lim_{x \rightarrow \infty} \frac{4}{x^2}$$

$$2 //$$

$$(14) \lim_{x \rightarrow 0} \frac{4x^3 + 2x^2 + x}{3x^2 + 2x}$$

$$(15) \lim_{x \rightarrow \infty} \frac{x+1}{x}$$

$$\lim_{x \rightarrow 0} \frac{\cancel{4x^3} + \cancel{2x^2} + x}{\cancel{3x^2} + \cancel{2x}}$$

$$\lim_{x \rightarrow \infty} \frac{x(1 + \frac{1}{x})}{x}$$

$$\lim_{x \rightarrow 0} (4x^2 - 2x + 1)$$

$$\lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} \frac{1}{x}$$

$$\lim_{x \rightarrow 0} (3x + 2)$$

$$1 + 0$$

$$1 //$$

$$\frac{1}{2} //$$

$$(v) \lim_{x \rightarrow \alpha} \frac{3x^2 - 2x - 1}{x^2 + 4}$$

$$\lim_{x \rightarrow \alpha} \frac{\cancel{x^2} \left(3 - \frac{2}{\cancel{x}} - \frac{1}{\cancel{x^2}} \right)}{\cancel{x^2} \left(1 + \frac{4}{\cancel{x^2}} \right)}$$

$$\lim_{x \rightarrow \alpha} 3 - \lim_{x \rightarrow \alpha} \frac{2}{x} + \lim_{x \rightarrow \alpha} \frac{1}{x^2}$$

$$\lim_{x \rightarrow \alpha} 1 + \lim_{x \rightarrow \alpha} \frac{4}{x^2}$$

$$\frac{3}{2} //$$

$$(vii) \lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x^2 - 12x + 20}$$

$$\lim_{x \rightarrow 2} \frac{\cancel{(x-2)} (x-3)}{\cancel{(x-2)} (x-10)}$$

$$\lim_{x \rightarrow 2} \frac{(x-3)}{(x-10)}$$

$$\lim_{x \rightarrow 2} \frac{(x-10)}{(x-10)}$$

$$\frac{-1}{-8} = \frac{1}{8}$$

$$(vi) \lim_{x \rightarrow 1} \frac{(x^3 - 1)}{(x^2 - 1)}$$

$$\lim_{x \rightarrow 1} \frac{\cancel{(x-1)} (x^2 + x + 1)}{\cancel{(x-1)} (x+1)}$$

$$\lim_{x \rightarrow 1} \frac{(x^2 + x + 1)}{(x+1)}$$

$$\lim_{x \rightarrow 1} (x+1)$$

$$\frac{3}{2} //$$

$$(viii) \lim_{x \rightarrow -2} \frac{x^3 + 3x^2 + 2x}{x^2 - x - 6}$$

$$\lim_{x \rightarrow -2} \frac{x(x^2 + 3x + 2)}{(x-3)(x+2)}$$

$$\lim_{x \rightarrow -2} \frac{\cancel{x} (x+2) (x+1)}{\cancel{(x+2)} (x-3) \cancel{(x+2)}}$$

$$\lim_{x \rightarrow -2} \frac{x+1}{x-3}$$

$$\lim_{x \rightarrow -2} \frac{(x-3)}{(x-3)}$$

$$\frac{-2+1}{-3} = \frac{-1}{-3} = \frac{1}{3} //$$

$$(ix) \lim_{x \rightarrow 1} \left[\frac{1}{1-x} - \frac{3}{(1-x^3)} \right]$$

$$\lim_{x \rightarrow 1} \left[-\frac{1}{(x-1)} + \frac{3}{x^3-1} \right]$$

$$\lim_{x \rightarrow 1} \left[\frac{3}{(x-1)(x^2+x-1)} - \frac{1}{(x-1)} \right]$$

$$\lim_{x \rightarrow 1} \left[\frac{3 - (x^2+x-1)}{(x-1)(x^2+x-1)} \right]$$

$$\lim_{x \rightarrow 1} \frac{3 - x^2 - x + 1}{(x-1)(x^2+x-1)}$$

$$\lim_{x \rightarrow 1} \frac{-x^2 - x + 2}{(x-1)(x^2+x-1)}$$

$$-1 \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{(x-1)(x^2+x-1)}$$

$$-1 \lim_{x \rightarrow 1} \frac{(x+2)(x-1)}{(x-1)(x^2+x-1)}$$

$$-1 \times \frac{3}{2}$$

$$-\frac{3}{2} //$$

~~(x)~~

$$(x) \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$$

$$\lim_{x \rightarrow 0} \frac{(\sqrt{1+x} - 1)(\sqrt{1+x} + 1)}{x(\sqrt{1+x} + 1)}$$

$$\lim_{x \rightarrow 0} \frac{1+x-1}{x(\sqrt{1+x} + 1)}$$

$$\lim_{x \rightarrow 0} \frac{1}{(\sqrt{1+x} + 1)}$$

$$\frac{1}{2} //$$

~~(xi)~~

$$(xi) \lim_{x \rightarrow 4} \frac{\sqrt{2x+1} - 3}{\sqrt{x-2} - \sqrt{2}}$$

$$\lim_{x \rightarrow 4} \frac{(\sqrt{2x+1} - 3)(\sqrt{x-2} + \sqrt{2})}{(\sqrt{x-2} - \sqrt{2})(\sqrt{x-2} + \sqrt{2})}$$

$$\lim_{x \rightarrow 4} \frac{(\sqrt{2x+1} - 3)(\sqrt{x-2} + \sqrt{2})}{x-2-2}$$

$$\lim_{x \rightarrow 4} \frac{(\sqrt{x-2} + \sqrt{2})(\sqrt{2x+1} - 3)(\sqrt{2x+1} + 3)}{(x-4)(\sqrt{2x+1} + 3)}$$

$$\lim_{x \rightarrow 4} \frac{(\sqrt{x-2} + \sqrt{2})(2x+1-9)}{(x-4)(\sqrt{2x+1} + 3)}$$

$$\lim_{x \rightarrow 4} \frac{(\sqrt{x-2} + \sqrt{2}) 2(x-4)}{(\sqrt{2x+1} + 3)(x-4)}$$

$$2 \lim_{x \rightarrow 4} (\sqrt{x-2} + \sqrt{2})$$

$$\lim_{x \rightarrow 4} (\sqrt{2x+1} + 3)$$

$$2 \times \sqrt{2}$$

$$2\sqrt{2}$$

$$= \frac{2\sqrt{2}}{3}$$

$$3 //$$

$$12. \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + p^2} - p}{\sqrt{x^2 + q^2} - q}$$

$$\lim_{x \rightarrow 0} \frac{(\sqrt{x^2 + p^2} - p)(\sqrt{x^2 + p^2} + p)(\sqrt{x^2 + q^2} + q)}{(\sqrt{x^2 + q^2} - q)(\sqrt{x^2 + q^2} + q)(\sqrt{x^2 + p^2} + p)}$$

$$\lim_{x \rightarrow 0} \frac{(x^2 + p^2 - p^2)(\sqrt{x^2 + q^2} + q)}{(x^2 + q^2 - q^2)(\sqrt{x^2 + p^2} + p)}$$

$$\lim_{x \rightarrow 0} \frac{(\sqrt{x^2 + q^2} + q)}{(\sqrt{x^2 + p^2} + p)}$$

$$\frac{2q}{2p} = \frac{q}{p} //$$

$$13. \lim_{x \rightarrow 1} \frac{3\sqrt{x} - 1}{\sqrt{x} - 1}$$

$$\lim_{x \rightarrow 1} \frac{(x)^{1/3} - 1^{1/3}}{(x)^{1/2} - 1^{1/2}}$$

$$\frac{\frac{1}{3}(1)^{-2/3}}{\frac{1}{2}(1)^{-1/2}} = \frac{\frac{1}{3}(1)^{-2/3} \times 1^{1/2}}{\frac{1}{2}(1)^{-1/2} \times 1^{2/3}} = \frac{2}{3} //$$

$$(14) \lim_{x \rightarrow a} \frac{\sqrt[n]{x} - \sqrt[n]{a}}{x - a}$$

$$\lim_{x \rightarrow a} \frac{(x)^{1/n} - (a)^{1/n}}{x - a}$$

$$\frac{1}{n} (a)^{1/n - 1}$$

$$\frac{1}{n} (a)^{\left(\frac{1-n}{n}\right)}$$

$$(15) \lim_{x \rightarrow a} \left(\sqrt{x^2 + ax} - \sqrt{x^2 - ax} \right)$$

$$\lim_{x \rightarrow a} \frac{(\sqrt{x^2 + ax} - \sqrt{x^2 - ax}) (\sqrt{x^2 + ax} + \sqrt{x^2 - ax})}{(\sqrt{x^2 + ax} + \sqrt{x^2 - ax})}$$

$$\lim_{x \rightarrow a} \frac{x^2 + ax - x^2 - ax}{(\sqrt{x^2 + ax} + \sqrt{x^2 - ax})}$$

$$\lim_{x \rightarrow a} \frac{0}{(\sqrt{x^2 + ax} + \sqrt{x^2 - ax})}$$

$$(15) \lim_{x \rightarrow 0} \left(\frac{\sqrt{1+x+x^2} - 1}{x} \right)$$

$$\lim_{x \rightarrow 0} \frac{(\sqrt{1+x+x^2} - 1) (\sqrt{1+x+x^2} + 1)}{x (\sqrt{1+x+x^2} + 1)}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x+x^2} - 1}{x (\sqrt{1+x+x^2} + 1)}$$

$$\lim_{x \rightarrow 0} \frac{x(x+1)}{x(\sqrt{1+x+x^2} + 1)}$$

$$\frac{1}{2} //$$

$$(16) \lim_{x \rightarrow a} \left(\sqrt{x^2 + ax} - \sqrt{x^2 - ax} \right)$$

$$\lim_{x \rightarrow a} \frac{(\sqrt{x^2 + ax} - \sqrt{x^2 - ax})(\sqrt{x^2 + ax} + \sqrt{x^2 - ax})}{\sqrt{x^2 + ax} + \sqrt{x^2 - ax}}$$

$$\lim_{x \rightarrow a} \frac{x^2 + ax - x^2 + ax}{(\sqrt{x^2 + ax} + \sqrt{x^2 - ax})}$$

$$2a \lim_{x \rightarrow a} \frac{x}{(\sqrt{x^2 + ax} + \sqrt{x^2 - ax})}$$

$$2a \lim_{x \rightarrow a} \frac{x}{(x\sqrt{1+a/x} + x\sqrt{1-a/x})}$$

$$2a \lim_{x \rightarrow a} \frac{1}{(\sqrt{1+a/x} + \sqrt{1-a/x})}$$

$$2a \lim_{x \rightarrow a} 1$$

$$\lim_{x \rightarrow a} (\sqrt{1+a/x}) + \lim_{x \rightarrow a} (\sqrt{1-a/x})$$

$$2a \times 1 = 2a //$$

$$(17) \lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 1}}{(x + 1)}$$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 \left(1 + \frac{1}{x^2}\right)}}{(x + 1)}$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{x} \sqrt{\left(1 + \frac{1}{x^2}\right)}}{\cancel{x} \left(1 + \frac{1}{x}\right)}$$

$$\lim_{x \rightarrow \infty} \frac{\sqrt{1 + \frac{1}{x^2}}}{\left(1 + \frac{1}{x}\right)}$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)$$

1/1/1

$$(18) \lim_{n \rightarrow \infty} \frac{1 + 2 + 3 + \dots + n}{n^2}$$

$$\sum_{n=2}^{\infty} \frac{1 + 2 + 3 + \dots + n}{2}$$

$$\lim_{n \rightarrow \infty} \frac{\frac{n(n+1)}{2}}{n^2}$$

$$\frac{1}{2} \lim_{n \rightarrow \infty} \frac{n(n+1)}{n^2}$$

$$\frac{1}{2} \lim_{n \rightarrow \infty} \frac{(n+1)}{n}$$

$$\frac{1}{2} \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)$$

$$\frac{1}{2} \left(\lim_{n \rightarrow \infty} 1 + \lim_{n \rightarrow \infty} \frac{1}{n} \right)$$

1/2 //

$$(19) \quad \lim_{n \rightarrow \infty} \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n^3}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2$$

$$\text{Sum} = \frac{n(n+1)(2n+1)}{6}$$

$$\lim_{n \rightarrow \infty} \frac{n(n+1)(2n+1)}{6n^3}$$

$$\lim_{n \rightarrow \infty} \frac{n(n+1)(2n+1)}{6n^3}$$

$$\lim_{n \rightarrow \infty} \frac{n^3 \left(1 + \frac{1}{n}\right) \left(2 + \frac{1}{n}\right)}{6n^3}$$

$$\frac{1}{6} \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right) \lim_{n \rightarrow \infty} \left(2 + \frac{1}{n}\right)$$

$$\frac{1}{6} \times 1 \times 2$$

$$\frac{1}{3} //$$

20) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^5 - 32}$

$\lim_{x \rightarrow 2} \frac{x^3 - 2^3}{x^5 - 2^5}$

$$\frac{3 \times (2)^2}{5 \times (2)^4} = \frac{3}{5 \times 2^2}$$

$$= \frac{3}{20} //$$

21) $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$

$$\lim_{x \rightarrow 0} \frac{1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots - 1}{x}$$

$$\lim_{x \rightarrow 0} \frac{\frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots}{x}$$

$$\lim_{x \rightarrow 0} \frac{1}{1!} + \frac{x}{2!} + \frac{x^2}{3!} + \dots$$

$$\lim_{x \rightarrow 0} \frac{1}{1!} = 1 //$$

$$(22) \lim_{x \rightarrow 0} \frac{a^x - 1}{x}$$

(i) $\lim_{x \rightarrow 0}$

$$y = a^x \text{ বিবেচনা}$$

$$\ln y = \ln a^x$$

$$\ln y = x \ln a$$

$$x = \frac{\ln y}{\ln a}$$

$$x \rightarrow 0 \text{ বিবেচনা } y = a^0 = 1$$

$$\lim_{y \rightarrow 1} \frac{y - 1}{\ln y}$$

$$\lim_{y \rightarrow 1} \frac{(y-1) \ln a}{\ln y}$$

$$\ln a \lim_{y \rightarrow 1} \frac{y-1}{\ln y}$$

$$\ln y = t = \log_e y$$

$$e^t = y$$

$$\lim_{y \rightarrow 1} \frac{y-1}{\ln y} = \lim_{t \rightarrow 0} \frac{e^t - 1}{t}$$

$$\lim_{y \rightarrow 1} = \lim_{e^t \rightarrow 1} = \lim_{t \rightarrow 0}$$

$$e^t = 1 + \cancel{t} + \frac{t}{1!} + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots$$

$$e^t - 1 = \cancel{1} + \frac{t}{1!} + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots \quad \cancel{1}$$

$$e^t - 1 = t \left[\frac{1}{1!} + \frac{t}{2!} + \frac{t^2}{3!} + \dots \right]$$

$$\frac{e^t - 1}{t} = \frac{1}{1!} + \frac{t}{2!} + \frac{t^2}{3!} + \dots$$

$$\lim_{t \rightarrow 0} \frac{e^t - 1}{t}$$

$$\ln a \left(\lim_{t \rightarrow 0} \frac{e^t - 1}{t} \right)$$

$$\ln a \times 1$$

$$\ln a$$

(ii) $\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$$

$$y = a^x - 1 \text{ થો.}$$

$$a^x = y + 1$$

છેલ્લે,

$$\ln a^x = \ln(y + 1)$$

$$x \ln a = \ln(y + 1)$$

$$x = \frac{\ln(y + 1)}{\ln a}$$

$$\ln a$$

$$y \rightarrow 0 \text{ થો } x \rightarrow 0$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$$

$$\lim_{y \rightarrow 0} \frac{y}{\ln(y + 1)}$$

$$\ln a$$

$$\lim_{y \rightarrow 0} \frac{y \ln a}{\ln(y + 1)}$$

$$\ln a \lim_{y \rightarrow 0} \frac{1}{\frac{1}{y} \ln(y + 1)}$$

$$\ln a \quad \lim_{y \rightarrow 0} \frac{1}{\ln(y+1)} y.$$

$$\lim_{y \rightarrow 0} \frac{\ln a}{\ln(y+1)^y y}$$

$$\ln a$$

$$\ln \left[\lim_{y \rightarrow 0} (y+1)^y y \right]$$

ඉහත ~~ස~~ $\lim_{y \rightarrow 0} (y+1)^y$ ආකෂේප කාණ්ඩය
 ද ආවේණිකව $\odot$ බව බව පෙන්විය.

$$\lim_{x \rightarrow 0} (x+1)^{1/x}$$

$$y = \lim_{x \rightarrow 0} (x+1)^{1/x}$$

$$\ln y = \lim_{x \rightarrow 0} \left[\lim_{x \rightarrow 0} (x+1)^{1/x} \right]$$

$$\ln y = \lim_{x \rightarrow 0} \left[\ln (x+1)^{1/x} \right]$$

$$\lim_{x \rightarrow 0} \left[\frac{1}{x} \ln(x+1) \right]$$

$$\ln y = \lim_{x \rightarrow 0} \left[\frac{\ln(x+1)}{x} \right]$$

$$\lim y = \lim_{x \rightarrow 0} \frac{\frac{d}{dx} \ln(1+x)}{\frac{d}{dx} x}$$

$$= \lim_{x \rightarrow 0} \frac{1}{1+x}$$

$$= \lim_{x \rightarrow 0} \frac{1}{1+x}$$

$$\ln y = 1$$

$$\ln y = 1$$

$$y = e$$

$$y = e$$

$$\therefore \text{সত্যি } \lim_{x \rightarrow 0} (x+1)^{\frac{1}{x}} = e$$

সত্যি $\ln a$

$$\ln \left[\lim_{y \rightarrow 0} (y+1)^{\frac{1}{y}} \right]$$

$$\boxed{\ln e = 1}$$

$$\frac{\ln a}{1} = \ln a = \ln a$$

$$(28) \lim_{x \rightarrow 1} \frac{x + x^2 + x^3 + \dots + x^n - n}{(x-1)}$$

$$\lim_{x \rightarrow 1} \frac{x + x^2 + x^3 + \dots + x^n - (1 + 1 + 1 + \dots + 1)}{(x-1)}$$

$$\lim_{x \rightarrow 1} \frac{(x-1) + (x^2-1) + (x^3-1) + \dots + (x^n-1)}{(x-1)}$$

$$\lim_{x \rightarrow 1} \frac{(x-1) [1 + (x+1) + (x^2+x+1) + \dots + x^{n-1}]}{(x-1)}$$

$$\lim_{x \rightarrow 1} \frac{(x^1-1)^1}{(x-1)} + \lim_{x \rightarrow 1} \frac{x^2-1^2}{x-1} + \lim_{x \rightarrow 1} \frac{x^3-1^3}{x-1} + \dots + \lim_{x \rightarrow 1} \frac{x^n-1^n}{x-1}$$

$$1 \times 1^0 + 2 \times 1^1 + 3 \times 1^2 + \dots + n \times 1^{n-1}$$

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1}$$

$$= 1 + 2 + 3 + 4 + \dots + n$$

$$= \frac{n(n+1)}{2}$$

24) $\lim_{x \rightarrow 1} \frac{3\sqrt{x^2} - 2\sqrt[3]{x} + 1}{(x-1)^2} \quad (\%) \times$

$\lim_{x \rightarrow 1} \frac{x^{2/3} - 2x^{1/3} + 1}{(x-1)^2}$

$\lim_{x \rightarrow 1} \frac{\frac{2}{3}(x)^{-1/3} - \frac{2}{3}(x)^{-2/3}}{2(x-1)} \quad (\%) \times$
 අලුත්
 2 නිවැරදි
 ප්‍රශ්න ලකුණ.

$\lim_{x \rightarrow 1} \frac{\frac{2}{3}x - \frac{1}{3}(x)^{-4/3} + \frac{4}{9}(x)^{-5/3}}{2}$
 අලුත්
 2 නිවැරදි
 ප්‍රශ්න ලකුණ.

$\lim_{x \rightarrow 1} \frac{\frac{2}{3}x - \frac{1}{3}(x)^{-4/3} + \lim_{x \rightarrow 1} \frac{4}{9}(x)^{-5/3}}{2}$

$\frac{-\frac{2}{9} + \frac{4}{9}}{2} = \frac{1}{9}$

25) $\lim_{x \rightarrow 0} \frac{(\sqrt{1+x} - \sqrt{1-x})}{x}$

$\lim_{x \rightarrow 0} \frac{(\sqrt{1+x} - \sqrt{1-x})(\sqrt{1+x} + \sqrt{1-x})}{x(\sqrt{1+x} + \sqrt{1-x})}$

$\lim_{x \rightarrow 0} \frac{\cancel{1+x} - \cancel{1-x}}{x(\sqrt{1+x} + \sqrt{1-x})}$

2 $\lim_{x \rightarrow 0} \frac{1}{x(\sqrt{1+x} + \sqrt{1-x})}$

2 x $\frac{1}{2} = 1$

26) $\lim_{x \rightarrow 64} \frac{x^{\frac{1}{2}} - 8}{x^{\frac{1}{3}} - 4}$

$\lim_{x \rightarrow 64} \frac{x^{\frac{1}{2}} - 8}{x^{\frac{1}{3}} - 4}$

$\frac{\frac{1}{2}(64)^{-\frac{1}{2}}}{\frac{1}{3}(64)^{-\frac{2}{3}}}$

$\frac{3}{2}$

27. $(1+m\alpha)^{1/3}$

$$\lim_{\alpha \rightarrow 0} \frac{(1+m\alpha)^{1/3} - 1}{\alpha}$$

$$m \lim_{\alpha \rightarrow 0} \frac{(1+m\alpha)^{1/3} - 1^{1/3}}{\alpha}$$

$$m \lim_{\alpha \rightarrow 0} \frac{(1+m\alpha)^{1/3} - 1^{1/3}}{\alpha}$$

$$m \times \frac{1}{3} (1)^{-2/3}$$

$$\frac{m}{3} //$$

28. $\lim_{x \rightarrow 0} \frac{x}{(1+2x)^{1/4} - 1}$

$$\frac{1}{2} \lim_{x \rightarrow 0} \frac{1+2x - 1}{(1+2x)^{1/4} - 1^{1/4}}$$

$$\frac{1}{2} \lim_{x \rightarrow 0} \frac{(1+2x) - 1}{(1+2x)^{1/4} - 1^{1/4}}$$

$$\frac{1}{2} \times \lim_{x \rightarrow 0} \frac{(1+2x)^{1/4} - 1^{1/4}}{(1+2x) - 1}$$

$$\frac{1}{2} \times \frac{1}{\frac{1}{4}(1)^{-3/4}}$$

$$\frac{1}{2} \times 4 = 2 //$$

29) $\lim_{x \rightarrow a} \frac{\sqrt{ax} - x}{x - a}$

$$\lim_{x \rightarrow a} \frac{(\sqrt{ax} - x)(\sqrt{ax} + x)}{(x - a)(\sqrt{ax} + x)}$$

$$\frac{(x - a)(\sqrt{ax} + x)}{(x - a)(\sqrt{ax} + x)}$$

$$\lim_{x \rightarrow a} \frac{ax - x^2}{(x - a)(\sqrt{ax} + x)}$$

$$\lim_{x \rightarrow a} \frac{-x(\cancel{x - a})}{(\cancel{x - a})(\sqrt{ax} + x)}$$

$$\frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{ax}}{(ax)^{1/2}}$$

$$\lim_{x \rightarrow 0} \frac{-x}{x(1 + \sqrt{a/x})} = \frac{-1}{1 + \sqrt{a/x}}$$

$$\sqrt{ax}$$

Asu

$$(30) \quad \lim_{x \rightarrow 9} \frac{9 - x^2}{(\sqrt{3x} - 3)}$$

$$\lim_{x \rightarrow 9} \frac{(3-x)(3+x)(\sqrt{3x}+3)}{(\sqrt{3x}-3)(\sqrt{3x}+3)}$$

$$\lim_{x \rightarrow 9} \frac{(3-x)(3+x)(\sqrt{3x}+3)}{3x-9}$$

$$\lim_{x \rightarrow 9} \frac{(3-x)(3+x)(\sqrt{3x}+3)}{3(x-3)}$$

$$= \lim_{x \rightarrow 9} \frac{\cancel{(x-3)}(3+x)(\sqrt{3x}+3)}{3\cancel{(x-3)}}$$

$$= \frac{1}{3} \lim_{x \rightarrow 9} (x+3) \lim_{x \rightarrow 9} (\sqrt{3x}+3)$$

$$= \frac{1}{3} \times 6 \times 6$$

$$= 12 //$$

3)

$$\lim_{x \rightarrow 1} \frac{x^2 - x^{-2}}{x^5 - x^{-5}}$$

$$\lim_{x \rightarrow 1} \frac{x^2 - \frac{1}{x^2}}{x^5 - \frac{1}{x^5}}$$

$$\lim_{x \rightarrow 1} \frac{\frac{x^4 - 1}{x^2}}{x^{10} - 1}$$

$$\lim_{x \rightarrow 1} \left(\frac{x^4 - 1}{x^2} \right) \times \frac{x^5 x^3}{(x^{10} - 1)}$$

$$\lim_{x \rightarrow 1} x^3$$

$$\lim_{x \rightarrow 1} \frac{(x^4 - 1)^4}{(x^{10} - 1)^4}$$

1 x

4 (1)³

10 (1)⁹

$\frac{4}{10}$